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sshd

Enables secure remote access to the machine over SSH with automatic host key management and optional CA-signed host certificates.

System

Network

What it does

- Generates and persists SSH host keys via `vars`.
- Optionally issues CA-signed host certificates for servers.
- Installs the `server` CA public key into `clients` `known_hosts` for TOFU-less verification.

When to use it

- Zero-TOFU SSH for dynamic fleets: admins/CI can connect to frequently rebuilt hosts (e.g., `server-1.example.com`) without prompts or per-host `known_hosts` churn.

Roles

- Server: runs `sshd`, presents a CA-signed host certificate for `<machine>.<domain>`.
- Client: trusts the CA for the given domains to verify servers' certificates. Tip: assign both roles to a machine if it should both present a cert and verify others.

Quick start (with host certificates) Useful if you never want to get a prompt about trusting the ssh fingerprint.

```

1 {
2   inventory.instances = {
3     sshd-with-certs = {
4       module = { name = "sshd"; input = "clan-core"; };
5       # Servers present certificates for <machine>.example.com
6       roles.server.tags.all = { };
7       roles.server.settings = {
8         certificate.searchDomains = [ "example.com" ];
9         # Optional: also add RSA host keys
10        # hostKeys.rsa.enable = true;
11      };
12      # Clients trust the CA for *.example.com
13      roles.client.tags.all = { };
14      roles.client.settings = {
15        certificate.searchDomains = [ "example.com" ];
16      };
17    };
18  };
19 }

```

Basic: only add persistent host keys (ed25519), no certificates Useful if you want to get an ssh "trust this server" prompt once and then never again.

```

1 {
2   inventory.instances = {
3     sshd-basic = {
4       module = {
5         name = "sshd";
6         input = "clan-core";
7       };
8       roles.server.tags.all = { };
9     };
10  };
11 }

```

Example: selective trust per environment Admins should trust only production; CI should trust prod and staging. Servers are reachable under both domains.

```

1 {
2   inventory.instances = {
3     sshd-env-scoped = {
4       module = { name = "sshd"; input = "clan-core"; };
5
6       # Servers present certs for both prod and staging FQDNs
7       roles.server.tags.all = { };

```

```
8     roles.server.settings = {
9         certificate.searchDomains = [ "prod.example.com" "staging.example.com" ];
10    };
11
12    # Admin laptop: trust prod only
13    roles.client.machines."admin-laptop".settings = {
14        certificate.searchDomains = [ "prod.example.com" ];
15    };
16
17    # CI runner: trust prod and staging
18    roles.client.machines."ci-runner-1".settings = {
19        certificate.searchDomains = [ "prod.example.com" "staging.example.com" ];
20    };
21 };
22 };
23 }
```

Explanation

- Admin -> server1.prod.example.com: zero-TOFU (verified via cert).
- Admin -> server1.staging.example.com: falls back to TOFU (or is blocked by policy).
- CI -> either prod or staging: zero-TOFU for both. Note: server and client searchDomains don't have to be identical; they only need to overlap for the hostnames you actually use.

Notes

- Connect using a name that matches a cert principal (e.g., `server1.example.com`); wildcards are not allowed inside the certificate.
- CA private key stays in `vars` (not deployed); only the CA public key is distributed.
- Logins still require your user SSH keys on the server (passwords are disabled).

Roles

The sshd service has the following roles:

- client
- server

Options for the `client` role

certificate.searchDomains

List of domains to include in the certificate. This option will prepend the machine name in front of each domain before adding it to the certificate.

Type: `list of string`

Default:

```
[ ]
```

Example

```
[  
  "mydomain.com"  
]
```

Declared in: clanServices/sshd/default.nix

Options for the `server` role

certificate.searchDomains

List of domains to include in the certificate. This option will prepend the machine name in front of each domain before adding it to the certificate.

Type: `list of string`

Default:

```
[ ]
```

 Example

```
[  
  "mydomain.com"  
]
```

Declared in: [clanServices/sshd/default.nix](#)

hostKeys.rsa.enable

Whether to enable generating a RSA host key.

Type:

Default:

false

 Example

true

Declared in: [clanServices/sshd/default.nix](#)

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